# Worksheet: Spreadsheets and Interest

**Group names:** _____

| problem statement | calculation |
|---|---|
| 41+5619 | 5660 |
| 36*987 | 35532 |
| 45^3 | 91125 |
| 12! | 479001600 |
| | |
| bill | 72.5 |
| tip% | 0.18 |
| tip amount | 13.05 |
| total | 85.55 |

1. Use a spreadsheet to calculate the following quantities:
   (a) $471 + 5619 =$ 5660
   (b) $36 * 987 =$ 35532
   (c) $45^3 =$ 91125 (Use syntax `=45^3`)
   (d) 12!= 479001600 (Use syntax `=fact(12)`)
   (e) Use a spreadsheet to calculate your total bill if the dinner cost \$72.50 and you want to give an 18% tip. \$85.55

2. Use a spreadsheet to calculate the following if you invested \$4500 for 10 years, assuming **simple interest**. (Use the process from the lecture notes.)
   (a) how much interest would you earn after 10 years if your interest rate was 2%? \$900
   (b) How much total money did you have at the end of 10 years? \$5400
   (c) How much interest would you earn after 10 years if your interest rate was 3%? \$1350
   (d) How much total money did you have at the end of 10 years? \$5850
   (e) How much **additional interest** did you earn in increasing the rate from 2% to 3%? \$450

With formulas

| | A | B | C | D | E |
|---|---|---|---|---|---|
| 1 | P | r | t | I = P*r*t | A = P+I |
| 2 | 4500 | 0.02 | 10 | 900 | 5400 |
| 3 | 4500 | 0.03 | 10 | 1350 | 5850 |
| | | | | difference? | 450 |

compute
= E3 - E2

With fill down

| t | P | r | I | A | r | I | A |
|---|---|---|---|---|---|---|---|
| 0 | $4,500.00 | 0.02 | | $4,500.00 | 0.03 | | $4,500.00 |
| 1 | | | $90.00 | $4,590.00 | | $135.00 | $4,635.00 |
| 2 | | | $90.00 | $4,680.00 | | $135.00 | $4,770.00 |
| 3 | | | $90.00 | $4,770.00 | | $135.00 | $4,905.00 |
| 4 | | | $90.00 | $4,860.00 | | $135.00 | $5,040.00 |
| 5 | | | $90.00 | $4,950.00 | | $135.00 | $5,175.00 |
| 6 | | | $90.00 | $5,040.00 | | $135.00 | $5,310.00 |
| 7 | | | $90.00 | $5,130.00 | | $135.00 | $5,445.00 |
| 8 | | | $90.00 | $5,220.00 | | $135.00 | $5,580.00 |
| 9 | | | $90.00 | $5,310.00 | | $135.00 | $5,715.00 |
| 10 | | | $90.00 | $5,400.00 | | $135.00 | $5,850.00 |
| | | | | | | | |
| difference | ? =H12-E12 | | | | | | |

= 450